



## Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

### 1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

|                                                                               |                                                                                                                  |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Imię i nazwisko / Name, surname                                               | Grzegorz Piotrowski                                                                                              |
| Dyscyplina kształcenia / Discipline of education                              | informatyka techniczna i telekomunikacja / information and communication technology                              |
| Wydział / Faculty                                                             | W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology                   |
| Katedra / Department                                                          | K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence                                       |
| Data rozpoczęcia kształcenia / Date of commencement of the doctoral education | 01.10.2025                                                                                                       |
| Nr ORCID                                                                      | 0009-0004-6013-0461                                                                                              |
| Promotor/ promotorzy / Supervisor(s)                                          | dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012 |
| Promotor pomocniczy / Assistant supervisor                                    | —                                                                                                                |

### 2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Poprawa zdolności generalizacji modeli uczenia głębokiego w transferze rzeczywistość–symulacja–rzeczywistość dla fizycznej sztucznej inteligencji / Improving the generalization of deep learning models in real-to-sim-to-real transfer for physical AI

### 3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

*A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:*

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

| Okres realizacji zadania (semestr kształcenia) /<br>Task implementation period (semester) | Zwięzły opis zadania / Brief description of the task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semestr 1                                                                                 | <ol style="list-style-type: none"> <li>1. Literature review in the scope of sim-to-real transfer, domain randomization and domain adaptation, neural scene reconstruction from real data (such as 3D Gaussian Splatting) and deep learning for robot navigation and manipulation.</li> <li>2. Identification of the most important research gaps in the generalization of deep learning models trained in simulations built from real data.</li> <li>3. Extending theoretical knowledge, including machine learning theory and learning under distribution shift, necessary for further work on the PhD thesis.</li> <li>4. Participation in the Doctoral School courses.</li> </ol> |
| Semestr 2                                                                                 | <ol style="list-style-type: none"> <li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li> <li>2. Initial pilot experiments on reconstructing real scenes with 3D Gaussian Splatting and importing them into a navigation simulator, with regard to the gaps identified during the literature review.</li> <li>3. Specifying the Individual Research Plan.</li> <li>4. Participation in the Doctoral School courses.</li> </ol>                                                                                                                                    |



|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semestr 3 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Preparation of the research environment for robot navigation, including real scene captures, 3D reconstruction, a navigation simulator built from the reconstructions, a dataset-based proxy measuring the reconstruction-fidelity gap, and computing resources.</li><li>3. Conducting research on which reconstruction errors of a digital twin, varying across a scene, determine the generalization of navigation models trained in it.</li></ol>                                                                                                                                                                                                                                                                                           |
| Semestr 4 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the research on the harmful reconstruction errors and on randomizing each region of a digital twin according to its reconstruction uncertainty and task relevance.</li><li>3. Writing a scientific article for a conference from the ministerial list (regarding the reconstruction errors that determine generalization), planned for submission to NeurIPS 2027 (200 points).</li><li>4. Starting the research on localizing the sim-to-real gap inside the model, with first results on the stage at which task information is lost on real inputs.</li><li>5. Preparation of the Preludium research grant proposal to the National Science Centre.</li><li>6. Participation in an international scientific conference.</li></ol> |
| Semestr 5 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the research on localizing the sim-to-real gap inside the model and on aligning representations at that stage.</li><li>3. Writing a scientific article for a conference from the ministerial list (regarding the localization of the sim-to-real gap inside deep models), planned for submission to ICML 2028 or CVPR 2028 (200 points).</li><li>4. Carrying out research on selecting real data under a limited budget to correct both the digital twin and the model.</li></ol>                                                                                                                                                                                                                                                    |



|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semestr 6 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the research on the selection of real data, and research on predicting whether the improvements transfer to unseen environments, confirmed on robotic manipulation in a controlled sim-to-sim setting with hidden physical parameters.</li><li>3. Writing a scientific article for a conference from the ministerial list (regarding the selection of real data under a limited budget and the prediction of transfer), planned for submission to NeurIPS 2028 (200 points).</li><li>4. Validation of the developed navigation methods on a real mobile robot, in cooperation with a robotics laboratory of Wrocław University of Science and Technology.</li><li>5. Research internship at a foreign research group working on robot learning and sim-to-real transfer.</li></ol> |
| Semestr 7 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the validation of the developed navigation methods on a real mobile robot.</li><li>3. Writing a journal article from the ministerial list summarizing the results on the generalization of the developed methods across environments, tasks and the real robot.</li><li>4. Participation in an international scientific conference.</li><li>5. Preparing the structure and sketch of the doctoral dissertation.</li></ol>                                                                                                                                                                                                                                                                                                                                                          |
| Semestr 8 | <ol style="list-style-type: none"><li>1. Completing and submitting the journal article.</li><li>2. Editing the doctoral dissertation, preparation of the final version of the document.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

#### 4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

*Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.*

##### September 2029

*(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /*

*The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)*

#### 5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

*Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral*



*dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).*

Deep learning has brought remarkable progress in computer vision and natural language processing, largely thanks to vast amounts of data collected from the web. Physical AI, i.e., AI in robots and other embodied systems that perceive and act in the real world, cannot rely on such data. Its models need interaction data, which are slow, expensive and sometimes unsafe to collect with real robots. Therefore, such models are commonly trained in simulation. Recently, digital twins, i.e., simulations built from real data, have become a promising alternative to hand-crafted simulators: neural scene reconstruction methods, such as 3D Gaussian Splatting, turn a short recording of a real place into a photorealistic digital twin in which a model can be trained at scale. This forms a real-to-sim-to-real loop, in which real data are turned into a simulation, a model is trained there, and the model is transferred back to reality.

However, the value of this loop depends on how well the trained models generalize. A model that performs well in simulation often fails in reality, since no simulation reproduces the real world exactly: appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. The resulting drop in performance, called the sim-to-real gap, is caused by the shift between the distributions of simulated and real data, which makes it a particular case of learning under distribution shift and a central obstacle to physical AI. The methods proposed so far have several limitations. Domain randomization, even when fitted to real data, varies a few global parameters regardless of where the simulation is inaccurate, and invariant representations may discard information needed for the task. Moreover, the few real data used for adaptation usually serve to correct either the model or the simulation, but not both. Finally, most methods are evaluated on a single task and a few scenes, and it is not known which properties of the digital twin and of the learned representations determine whether a model generalizes to reality.

Therefore, the dissertation aims to improve the generalization of deep learning models in real-to-sim-to-real transfer at the points of the loop where the sim-to-real gap arises. The research shall focus on understanding which reconstruction errors of a digital twin are harmful to generalization, on localizing where inside a model the gap arises and reducing it there, on choosing the small amount of real data that corrects both the digital twin and the model most efficiently, and on verifying whether the obtained improvements carry over to unseen environments and to another physical task, and which properties of the shift explain when they do. Robot navigation shall serve as the main task, and robotic manipulation, in a controlled simulated setting, as a confirmation test. Most experiments shall use public recordings of real scenes, in which a second, independent capture measures the reconstruction-fidelity gap, while a real robot shall provide the main evidence for differences in actuation and sensors. Beyond more reliable models, the research could help explain why models trained on synthetic data succeed or fail on real data, which is relevant to machine learning in general.

The results have potential applications in service, logistics and assistive robotics, where a robot could be trained in a digital copy of a new warehouse, hospital or home and operate there reliably after little real-world adaptation. More broadly, they concern any domain in which models are trained on simulated or reconstructed data and deployed on real sensor data, such as autonomous driving or inspection robots.

## **6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation**

*Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including*



*references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*

Deep learning models for navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. Yet performance in simulation can poorly predict real-world performance unless the simulator is carefully tuned [1]. The sim-to-real gap is an instance of learning under distribution shift: domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions, and the joint error of the best single model on both domains [2]. Used as motivation only, the bound suggests three levers: a training distribution that covers reality, a model insensitive to the remaining discrepancy, and, since invariance cannot reduce the joint error, a simulation corrected towards reality.

The first lever is exploited by domain randomization, which varies simulator parameters such as textures so that reality appears as one more variation [3]. Since overly wide randomization leads to conservative behaviour, later works shape its distribution: they fit it to a few real rollouts [4] or maximize its entropy while preserving task success [5]. Yet they act on a few global parameters of a synthetic simulator, not on errors that vary across a scene.

The second lever is invariance. Domain-adversarial training aligns features across domains [6]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies, i.e., models that map observations to actions, in reality [7]. However, invariance with low source error does not guarantee transfer, as it can increase the joint error term [8], and full invariance may discard task information. Effective sim-and-real co-training aligns the two domains while keeping them distinguishable [9], so domain separability alone does not show where the gap harms the task. Moreover, invariance is usually enforced at a fixed location, the input or the final features.

A recent development is building digital twins. 3D Gaussian Splatting [10] reconstructs photorealistic scenes from ordinary images, and twins have been used to train manipulation policies from a scan of the target scene [11] and navigation policies from a phone capture of a room [12]. In twins, uncertainty has so far been modelled mainly for physical parameters [13]. Which simulation factors matter has been studied mainly globally: a large real-world manipulation study compared randomization, photorealistic rendering and physics realism [14], and in robotic ultrasound, the fidelity of task-relevant feature changes predicted transfer better than single-frame realism [15]. It remains open which reconstruction errors that vary across a scene harm generalization, and how training variation should follow the reconstruction uncertainty of each region.

It is also unclear where inside a model the gap arises. Manipulation encoders degrade under lighting and texture shifts [16]. Probing, i.e., reading out information from hidden states with simple classifiers, showed that action fine-tuning degrades the visual representations of robot policies [17], and the best layers to adapt depend on the type of shift [18]. However, the stage at which task information available in simulation stops being recoverable from real inputs has not been located in models trained in twins.

The remaining gap is usually reduced with little real data. Co-training with a small real set yields large gains [19], and active system identification collects the real trajectories most informative about physics [20]. In vision, selecting images to label by domain discrepancy and uncertainty beats random selection [21]. The closest work uses a twin to find failure-prone configurations for real rollouts [22]. However, these methods correct either the model or the simulation; correcting both from the same selected data, guided by the twin's reconstruction uncertainty, has, to our knowledge, not been studied.

Finally, the gap can be decomposed into factors of variation whose difficulty ordering is largely consistent between simulation and reality [23]. However, it is rarely tested whether measurements taken before a method is applied predict, better than simple indicators such as the size of the gap, whether its benefit carries over to a new scene or task.



These open problems motivate the research questions in Section 7.

## References

- [1] Kadian, A., et al. (2020). Sim2Real predictivity: Does evaluation in simulation predict real-world performance? RA-L.
- [2] Ben-David, S., et al. (2010). A theory of learning from different domains. Machine Learning.
- [3] Tobin, J., et al. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. IROS.
- [4] Chebotar, Y., et al. (2019). Closing the sim-to-real loop: Adapting simulation randomization with real world experience. ICRA.
- [5] Tiboni, G., et al. (2024). Domain randomization via entropy maximization. ICLR.
- [6] Ganin, Y., et al. (2016). Domain-adversarial training of neural networks. JMLR.
- [7] Cheng, S., et al. (2025). Generalizable domain adaptation for sim-and-real policy co-training. NeurIPS.
- [8] Zhao, H., et al. (2019). On learning invariant representations for domain adaptation. ICML.
- [9] Lei, Y., et al. (2026). A Mechanistic Analysis of Sim-and-Real Co-Training in Generative Robot Policies. arXiv:2604.13645.
- [10] Kerbl, B., et al. (2023). 3D Gaussian splatting for real-time radiance field rendering. ACM TOG.
- [11] Torne, M., et al. (2024). Reconciling reality through simulation: A real-to-sim-to-real approach for robust manipulation. RSS.
- [12] Chhablani, G., et al. (2025). EmbodiedSplat: Personalized real-to-sim-to-real navigation with Gaussian splats from a mobile device. ICCV.
- [13] Wang, M., et al. (2026). Phys2Real: Fusing VLM priors with interactive online adaptation for uncertainty-aware sim-to-real manipulation. ICRA.
- [14] Jin, R., et al. (2026). Grounding Sim-to-Real Generalization in Robotic Manipulation: An Empirical Study with Vision-Language-Action Models. arXiv:2603.22876.
- [15] Qian, Y., et al. (2026). Task-Relevant Feature-Dynamics Fidelity Enables Zero-Shot Sim-to-Real Transfer for Robotic Ultrasound Scanning. arXiv:2608.29516.
- [16] Burns, K., et al. (2024). What makes pre-trained visual representations successful for robust manipulation? CoRL.
- [17] Kachaev, N., et al. (2026). Don't blind your VLA: Aligning visual representations for OOD generalization. AAMAS.
- [18] Lee, Y., et al. (2023). Surgical fine-tuning improves adaptation to distribution shifts. ICLR.
- [19] Maddukuri, A., et al. (2025). Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. RSS.
- [20] Memmel, M., et al. (2024). ASID: Active exploration for system identification in robotic manipulation. ICLR.
- [21] Su, J.-C., et al. (2020). Active adversarial domain adaptation. WACV.
- [22] Xu, Q., et al. (2026). TwinRL: Digital twin-driven reinforcement learning for real-world robotic manipulation. arXiv:2602.09023.
- [23] Xie, A., et al. (2024). Decomposing the generalization gap in imitation learning for visual robotic manipulation. ICRA.

## 7. Pytania i hipotezy badawcze / Research questions and hypotheses

*Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).*

The research carried out in the proposed doctoral dissertation is aimed at studying the following research questions.

1. **Which reconstruction errors of a digital twin, varying across a scene, determine how well models trained in it generalize?** Domain adaptation theory, used here as motivation, bounds the real-world error of a model by its error in simulation, the discrepancy between the simulated and



real distributions, and a joint error term. A digital twin reproduces appearance, geometry and lighting only approximately, and its errors differ between regions of a scene. *Hypothesis*: The harm caused by a reconstruction error is predicted better by the change it induces in the representation of a fixed reference model on held-out paired views than by its magnitude in the image or geometry. Separately, randomizing each region in proportion to its reconstruction uncertainty and task relevance yields better generalization than uniform randomization. The research will replace components of the twin with a more accurate reference and measure the effect. If harm follows error magnitude, this attribution will itself be reported, and the third question will rely on reconstruction uncertainty and action sensitivity only.

2. **Where inside a model trained in simulation does the sim-to-real gap arise, and can aligning representations at that stage reduce it?** Models trained in simulation can rely on features absent in reality, such as reconstruction artifacts. Simulated and real inputs remain distinguishable even in models that transfer well, so the gap should be localized by what the model loses about the task, not by whether the domains can be told apart. *Hypothesis*: There is a first stage of the model at which task information that can be decoded from simulated inputs can no longer be decoded from real ones. Aligning simulated and real representations at that stage, from a few paired views, yields better generalization than aligning them at the input, at the final features or everywhere. The research will use probing of hidden states to find this stage before intervening.
3. **Which real data, chosen under a limited budget, most efficiently correct both the simulation and the model?** Real data are expensive, and existing methods use them to correct either the model or the simulation. However, the joint error term cannot be reduced by invariance, only by making the simulation closer to reality. *Hypothesis*: Correcting both the digital twin and the model from the same selected real data reduces the gap more than correcting either one alone. The data should be selected where the twin is predicted to disagree with reality in task-relevant ways. This disagreement can be estimated from the reconstruction uncertainty, the representation distance between simulated and real views, and the sensitivity of the predicted action. Such selection is expected to beat random and failure-driven selection. The research will compare selection rules at equal budgets, counting unlabelled real images as real data.
4. **Do the improvements generalize to unseen environments and to another physical task, and which properties of the shift explain when they do?** A valuable method should not be tied to a single scene or task. Therefore, the developed methods shall be applied with unchanged settings to many navigation scenes held out from training, and to robotic manipulation as a confirmation test in a controlled sim-to-sim setting with hidden physical parameters. *Hypothesis*: The attribution of the gap from the first question and its localization from the second can be measured in a new setting before a method is applied. These measurements predict whether the improvement transfers. They predict it better than simple indicators, such as the raw size of the gap or the image-level discrepancy. The research will test this prediction across held-out scenes and confirm it on manipulation.

## 8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

*Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).*

The proposed PhD dissertation aims to introduce several contributions to the development of the scientific discipline. The central scientific problem addressed is explaining and reducing the sim-to-real gap caused by the distribution shift between a digital twin and reality; poor generalization of models trained in simulation remains one of the main obstacles to applying machine learning in physical AI. The central claim of the dissertation is that this gap is concentrated in a small set of task-



relevant discrepancies, which can be identified among the reconstruction errors of the digital twin, localized at the stage of the model where task information is lost, and reduced by correcting both the digital twin and the model from the same small set of real data. Addressing them there is expected to yield better generalization than uniform randomization, alignment at a fixed location of the model, or real data chosen at random or at the model's failures. Research questions 1-3 test the three parts of this claim, and question 4 tests whether it holds across environments and tasks; the methods derived from it are intended to be general, i.e., not tied to a particular robot, scene or task, and will be examined mainly on robot navigation, with robotic manipulation as a controlled confirmation test.

Furthermore, the research is expected to deliver new methods and findings that do not exist at present: an attribution of the sim-to-real gap to reconstruction errors that vary across a scene, with a principle for randomizing each region of a digital twin according to its reconstruction uncertainty and task relevance; a method for localizing the gap as the first stage of a deep model at which task information is lost on real inputs, and for aligning representations only there; a rule for selecting real data that corrects the digital twin and the model at the same time; and an empirical account of whether properties of the shift, measured before a method is applied, predict better than simple indicators when these improvements carry over across environments and tasks. Such knowledge could complement the currently dominant randomization heuristics with more principled and interpretable approaches, and could introduce new perspectives for the design of simulations and of training procedures.

Since transfer from simulation to reality is an instance of learning under distribution shift, the findings are also expected to be relevant to machine learning in general. The analysis of which errors in the training distribution are harmful concerns domain adaptation; the localization of the sim-to-real gap inside a model extends probing methods developed for language and vision models and contributes to representation learning; and the selection of informative data under a budget concerns active learning and the adaptation of pretrained foundation models to new domains from a few examples.

Finally, the developed methods could reduce the amount of real data, and thus the cost, needed to deploy learning-based robots in new environments. The source code and the experimental setups are planned to be released, which could facilitate further research in the field and attract other researchers. Apart from purely scientific results, the methods could be used in real-world applications such as service, logistics and inspection robotics, as well as in other domains where models are trained in simulation, e.g., autonomous driving.

## 9. Planowane metody badawcze / Planned research methods

*Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*

The research will follow the methods commonly adopted in machine learning and robot learning projects. Since no established theory guarantees generalization from simulation to reality, the research will be mainly empirical. Domain adaptation theory, which relates the error of a model in reality to its error in simulation, the discrepancy between the two distributions and a joint error term, will serve as motivation, and the properties of the proposed methods will be analysed theoretically whenever possible. Numbers in brackets refer to the references listed in Section 6.

Digital twins will be built from real captures of indoor scenes, using structure-from-motion for camera poses and neural scene reconstruction, such as 3D Gaussian Splatting, for appearance and geometry, and then coupled with a navigation simulator, e.g., Habitat, with geometry for collisions and appearance rendered from the reconstruction. Reconstruction uncertainty will be estimated per region, e.g., from view coverage and the disagreement of an ensemble of reconstructions. Models for goal-directed indoor navigation (e.g., point-goal or image-goal) will be trained with reinforcement and imitation learning, starting from pretrained models when possible.



Experiments will be conducted mainly on public datasets of real scenes, such as ScanNet++ and MuSHRoom, which contain separate captures of the same scenes made with different devices. This allows building the digital twin from one capture and using another, independent capture as a reference. Real frames of the independent capture will serve for open-loop evaluation, in which the model does not act, measuring the discrepancy between representations of paired simulated and real views, the agreement of predicted actions, and probing accuracy. A reference simulation built from the independent, higher-fidelity capture (e.g., the laser scan and DSLR images in ScanNet++) will serve for closed-loop evaluation, in which the model's actions determine what it observes next. The difference between the two simulations measures the reconstruction-fidelity gap on many scenes without a robot, but not differences in actuation and sensors.

For the first question, variants of the digital twin will be built in which one component of a region, i.e., its appearance, geometry or lighting, is replaced by its counterpart from the reference reconstruction, and the resulting change of the gap will attribute it to types of reconstruction error. The harm of each error will be related to two predictors: its magnitude in the image or geometry, and the change it induces in the representation of a fixed reference model, e.g., a pretrained visual encoder, on held-out paired views. If harm follows error magnitude, this attribution will be reported as a result, and the third question will then use only reconstruction uncertainty and action sensitivity. As a separate claim, the training distribution will be shaped by randomizing appearance (colour and opacity of Gaussians), geometry and, where the representation allows relighting, lighting per region in proportion to its reconstruction uncertainty and task relevance, the latter estimated from the sensitivity of the model's actions to the region, e.g., by occluding it. This will be compared with uniform randomization, randomization fitted to real rollouts [4] and entropy-maximizing randomization [5].

For the second question, probes, i.e., simple classifiers, will be trained at each layer on simulated hidden states to decode task variables, such as the direction and distance to the goal or the presence of obstacles, and tested on the paired real views. The first stage at which task information decodable in simulation can no longer be decoded from real inputs will localize the gap. Unlike the separability of the two domains, which can remain high even in models that transfer well [9], this criterion refers to the task. An alignment loss (e.g., distribution-discrepancy or contrastive) will then be applied at that stage to a few paired simulated and real views at the same camera poses, and compared with alignment at the input, domain-adversarial alignment of the final features [6], alignment at all layers and full fine-tuning.

For the third question, the budget will count all real data: unlabelled real images as well as interaction data, such as robot trials or demonstrations. On the dataset-based proxy, a unit of interaction data will be an expert or policy episode executed in the reference simulation, paired with the real frames at the visited poses; on the physical robot, a real trial or demonstration. Real data will be selected by scores combining reconstruction uncertainty, the distance between representations of simulated and real views, and the sensitivity of the predicted action. The same selected data will be used to refine the digital twin, by re-optimizing the reconstruction in uncertain regions, and to adapt the model, by fine-tuning or co-training on simulated and real data. This will be compared with random and failure-driven selection [22], with the same score without task relevance, and with correcting only the model or only the digital twin, at equal real-data budgets. Data used for final evaluation will be disjoint from those available for selection.

For the fourth question, the methods will be applied with unchanged settings to many held-out navigation scenes. The attribution and localization measured in each scene before a method is applied will be related to the size of the improvement there, e.g., by rank correlation, and compared with simple predictors: the raw size of the gap and the image-level discrepancy between simulated and real views. The relation will then be confirmed on tabletop manipulation in a controlled sim-to-sim setting, e.g., in ManiSkill3, in which a simulation whose visual and physical parameters are hidden from the learner stands in for reality.



The quantitative evaluation will rely on standard metrics for embodied tasks, such as success rate and success weighted by path length (SPL), and on the sim-to-real gap, defined at two levels: on the dataset-based proxy, as the difference between the performance in the digital twin and in the reference simulation, complemented by the open-loop measures on real frames; and on the physical robot, as the difference between the closed-loop performance in simulation and in reality. The agreement between the two levels will be reported as a result, e.g., by the rank correlation of model scores. All methods will be compared with baselines under equal training effort and an equal amount of real data. Statistical methods will be leveraged to provide reliable outcomes, including repeated experiments with different random seeds and scenes, confidence intervals, and statistical tests for comparisons with baseline methods. According to accepted standards, the data used for training, model selection and final evaluation will be kept separate.

The navigation results will be validated in one block in semesters 6-7 on a wheeled mobile robot with an RGB-D camera, in cooperation with a robotics laboratory of the university. The validation will measure closed-loop success rate and SPL in reality and provide the main evidence for differences in actuation and sensors. Should robot access be limited, the conclusions will rest on the dataset-based proxy and a reduced set of real trials.

Experiments will be implemented mainly in Python with the PyTorch deep learning library. Code, datasets, configurations and results will be versioned with Git and experiment tracking tools for reproducibility. Scene reconstruction and policy training will run on GPU clusters of the Department and on the Polish computing infrastructure (WCSS and PLGrid). Code and data will be made publicly available whenever possible, following open science standards and respecting the licences of the datasets used.

The findings will be published at leading international machine learning, computer vision and robotics conferences, such as NeurIPS, ICML, ICLR, CVPR and RSS, and in journals from the ministerial list. Results will also be presented at smaller conferences and workshops, such as MLinPL, and at summer schools.

## 10. Streszczenie popularnonaukowe / Abstract for general public

*Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1).* / *The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

### Streszczenie popularnonaukowe

Sztuczna inteligencja coraz częściej wychodzi poza ekrany komputerów i trafia do systemów fizycznych, takich jak roboty poruszające się po budynkach lub przenoszące przedmioty. Modele uczenia głębokiego sterujące takimi robotami potrzebują ogromnej liczby przykładów, których zebranie w prawdziwym świecie jest powolne, kosztowne, a czasem niebezpieczne. Dlatego modele te uczy się zwykle w symulacji, a coraz częściej w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań. Taka kopia nigdy nie jest jednak idealna. Na przykład szklane drzwi lub odbicie w lustrze mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. Pozostaje to jedną z głównych przeszkód na drodze do szerszego zastosowania fizycznej sztucznej inteligencji. Celem rozprawy jest poprawa zdolności generalizacji modeli uczenia głębokiego trenowanych w cyfrowych bliźniakach, czyli umiejętności poprawnego działania w warunkach innych niż te, w których je trenowano, a zwłaszcza w rzeczywistości. Badania w pierwszej kolejności pozwolą ustalić, które różnice między symulacją a



rzeczywistością faktycznie szkodzą modelowi i na którym etapie przetwarzania w modelu przekształcają się one w błędy. Na tej podstawie zostaną opracowane metody, które nauczą model pomijać nieistotne różnice i więcej ćwiczyć tam, gdzie cyfrowy bliźniak jest najmniej dokładny, a także metody wyboru nielicznych rzeczywistych przykładów, które najlepiej poprawiają zarówno model, jak i cyfrowego bliźniaka. Metody zostaną sprawdzone przede wszystkim w zadaniu nawigacji robotów, w środowiskach, które nie były wykorzystywane podczas uczenia, i zweryfikowane na prawdziwym robocie mobilnym. Zostaną też przetestowane w symulowanym zadaniu manipulacji robotycznej. Oczekuje się, że rozprawa przyniesie lepsze zrozumienie przyczyn, dla których modele uczone w symulacji zawodzą w rzeczywistości, a także metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach i przy więcej niż jednym zadaniu.

### **Abstract for general public**

Artificial intelligence is increasingly moving beyond computer screens into physical systems, such as robots that move around buildings or carry objects. Deep learning models that control such robots need a huge number of examples, which are slow, costly and sometimes unsafe to collect in the real world. Therefore, such models are usually trained in simulation, and increasingly in digital twins, that is, virtual copies of real places rebuilt from recordings of those places. Such a copy is never perfect, however. For example, a glass door or a reflection in a mirror may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. This remains one of the main obstacles to the wider use of physical artificial intelligence. The goal of the dissertation is to improve the generalization of deep learning models trained in digital twins, i.e., their ability to work correctly in conditions different from those they were trained in, and in particular in reality. The research will first establish which differences between simulation and reality actually harm the model, and at which stage of processing inside the model they turn into errors. On this basis, methods will be developed that teach the model to ignore irrelevant differences and to practise more where the digital twin is least accurate, together with methods that choose the few real examples that best correct both the model and the digital twin. The methods will be examined mainly on robot navigation, in environments that were not used during training, and verified on a real mobile robot. They will also be tested on a simulated robotic manipulation task. The dissertation is expected to bring a better understanding of why models trained in simulation fail in reality, as well as methods that allow robots to learn to work in new places and on more than one task faster, more cheaply and more safely.

## **11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography**

*Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.*



May 2027 (planned submission to the NeurIPS 2027 conference, 200 points on the ministerial list).

## 12. Inne / Other comments

*Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).*

## 13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

*Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.*

It is planned to have weekly meetings with the supervisor. The meetings are planned to consist of results presentations, discussions, reviews and joint work on papers. The scope of meetings is also supposed to include the evaluation of progress and the planning of further activities. Moreover, it is planned to discuss results, issues and recent advancements in the field with other Ph.D. students weekly, with the supervisor's participation. Day-to-day communication will take place by e-mail and instant messaging.

## 14. Opinia promotora pomocniczego / Opinion of assistant supervisor

*Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.*

nie dotyczy / not applicable

.....  
Podpis promotora pomocniczego /  
Signature of assistant supervisor

## 15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....  
Data / Date

.....  
Podpis doktoranta / Signature of doctoral student

.....  
Podpis promotora / Signature of supervisor

.....  
Podpis drugiego promotora / Signature of second supervisor